

MATTER IN OUR SURROUNDINGS

❖ **Matter**

- Anything that occupies space and has mass is called matter.

❖ **Characteristics of particles of matter**

- Particles of matter have space between them.

We know that salt can be dissolved in water i.e. the particles of water molecules have space between them where the salt particles fit.

- Particles of matter are continuously moving.

We know that the particles in motion possess kinetic energy which is dependent on temperature. As the temperature increases, particles move faster and hence kinetic energy increases.

- Particles of matter attract each other.

There is a force of attraction between the particles which varies from matter to matter.

❖ **States of matter**

Matter can be classified as solid, liquid or gas on basis of its physical state.

- Solids

- (i) The molecules are closely packed together i.e. intermolecular distances are less.
- (ii) There is strong force of attraction between the molecules.
- (iii) Solids have definite shape and volume.
- (iv) Solids are rigid i.e. they can't be compressed.
- (v) Solids do not flow.

- Liquids

- (i) The molecules are loosely packed i.e. intermolecular distance is more than that of solids.
- (ii) They do not have strong force of attraction between molecules to hold them at fix position.
- (iii) Liquids have definite volume but do not have fixed shape.
- (iv) Liquids are slightly compressible.
- (v) Liquids can flow easily.

- Gases

- (i) The molecules are far apart from each other i.e. intermolecular distance is very large.
- (ii) They have weak force of attraction which can be approximated as negligible.
- (iii) Gases do not have definite shape and volume.
- (iv) Gases are highly compressible as they have large distance between their molecules which can be reduced by applying pressure. E.g. CNG(Compressed Natural Gas).
- (v) Gases can flow in any direction.

❖ **Diffusion**

- The particles of matter intermix with each other on their own. This intermixing of particles of two different types of matter on their own is known as diffusion.

- It is temperature dependent. i.e. on heating, diffusion becomes faster.
- The rate of diffusion is fastest in gases, moderate in liquids and slowest in solids due to the intermolecular distances.
- The smell of hot cooked food reaches us in seconds because the particles of aroma of food mix with the particles of air spread from the kitchen and reaches us.

❖ **Inter – conversion of state**

- The solid, liquid and gaseous states of a matter can be converted to each other by changing the temperature and pressure. E.g. water exists in solid state as ice, in liquid state as water and in gaseous state as steam.
- The kinetic energy of the particles increases if we increase the temperature of the solids. Due to this, particles start vibrating with greater speed. The energy supplied by heat becomes greater than the force of attraction between the particles which leads to free movement of the particles as they leave their fixed positions and there is a conversion to liquid state from solid state.

Melting point

- During inter-conversion of state, the solid melts and is converted into liquid state. The temperature at which solid melts to become liquid at atmospheric pressure is known as its melting point.
- If we continuously provide heat to the system when it is converted to liquid state, the particles start moving with greater speed and the intermolecular force decreases leading to movement of particles more freely and there is a conversion from liquid to gaseous state.

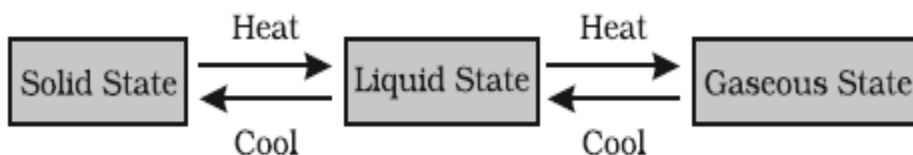
Boiling point

The temperature at which liquid starts boiling to become gas at atmospheric pressure is known as boiling point.

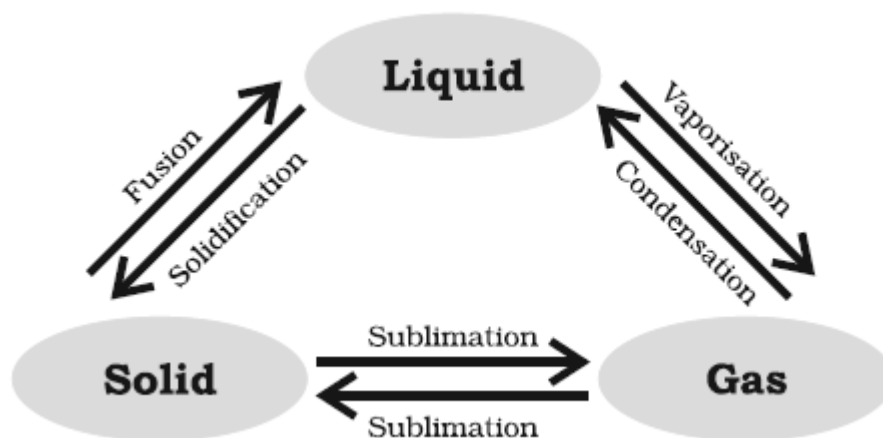
Latent heat of fusion

- Whenever inter – conversion of state takes place, then the temperature of the system does not change. This is because the energy supplied by heat is entirely used up in conversion of one state to another. This is known as latent heat or hidden heat.
- The amount of heat energy required to change 1 kg of a solid into liquid at atmospheric pressure at its melting point is called latent heat of fusion.
- Similarly, the amount of heat energy required to change 1 kg of a liquid at atmospheric pressure at its boiling point is called latent heat of vaporization.

So, we can conclude that the state of matter can be changed into another state by changing the temperature.



- The solid state can also be changed directly to gaseous state and vice – versa without changing into liquid state and is known as sublimation.
- Solid CO₂ gets converted directly to gaseous state on decreasing the pressure to 1 atmosphere without coming into liquid state. That is why solid CO₂ is known as dry ice.



- Conversion of solid state to liquid state is called fusion.
- Conversion of liquid state to gaseous state is called vaporization.
- Conversion of gaseous to liquid state is called condensation.
- Conversion of liquid to solid state is called solidification.
- Conversion of solid to gaseous or gaseous to solid state is called sublimation.

❖ 5 states of matter

- Nowadays, scientists are talking about five states of matter i.e. solid, liquid, gas, plasma and Bose – Einstein condensate.

Plasma

The state consists of super energetic and super excited particles. These particles are in the form of ionized gases. The fluorescent tube and neon sign bulbs consist of plasma. Inside a neon sign bulb, there is Neon gas and inside the fluorescent tube, there is Helium gas or some other gas. The gas gets ionized, i.e. gets charged when electrical energy flows through it. This charging up creates a plasma glowing inside the tube or bulb. The plasma glows with a special colour depending on the nature of gas. The sun and the stars glow because of the presence of plasma in them. The plasma is created in stars because of very high temperature.

Bose – Einstein Condensate

In 1920, Indian physicist Satyendra Nath Bose had done some calculation for fifth state of matter. Building on his calculations, Albert Einstein predicted new state of matter i.e. Bose – Einstein Condensate (BEC). In 2001, Eric A. Cornell, Wolfgang Ketterle and Carl E. Wieman of USA received the nobel prize in physics for achieving BEC. The BEC

is formed by cooling a gas of extremely low density, about one-hundred-thousandth the density of normal air, to super low temperatures.

❖ **Evaporation**

- In solid, liquid or gases, the particles move with different velocities or the particles have different kinetic energies.
- In liquid state, the particles on the surface have higher kinetic energy and can break the intermolecular forces to go in vapour phase.
- This phenomenon of change of a liquid into vapours at any temperature below its boiling point is called evaporation.

Factors affecting evaporation

(i) Surface area

If the surface area is increased, rate of evaporation increases. That is why we spread the clothes for drying.

(ii) Temperature

More the temperature more will be the kinetic energy of particle and hence more number of particles will go in vapour phase leading to increase in rate of evaporation.

(iii) Humidity

Humidity is the presence of water vapours in air. If the amount of water in air is already high, it decreases the rate of evaporation.

(iv) Wind speed

With increase in wind speed, the particles of water vapour move away with the wind, decreasing the amount of water vapour in surrounding and hence rate of evaporation increases.

Cooling effect of evaporation

- The particles of liquid absorb energy from the surrounding to regain the energy lost during evaporation. This absorption of energy from surroundings makes the surroundings cold.
- Some of the daily life examples are given below
 - (i) When we put some petrol or acetone (nail polish remover) on our palm, the particles gain energy from our palm or surroundings and evaporate causing the palm to feel cold.
 - (ii) After a hot sunny day, we sprinkle water on roof or open ground because the large latent heat of vaporization of water helps to cool the hot surface.
 - (iii) We wear cotton clothes during summer because cotton is a good absorber of water. So, it absorbs the sweat and exposes it to atmosphere for easy evaporation so that our body becomes cool.
 - (iv) Water is kept in earthen pot to cool the water inside it as the pot is porous leading to easy evaporation.
 - (v) We see water droplets in the surface of a glass containing ice-cold water because the water vapours present in air on coming in contact with cold glass of water loses energy and converted into liquid state which we see as water droplets.